Variable Drift

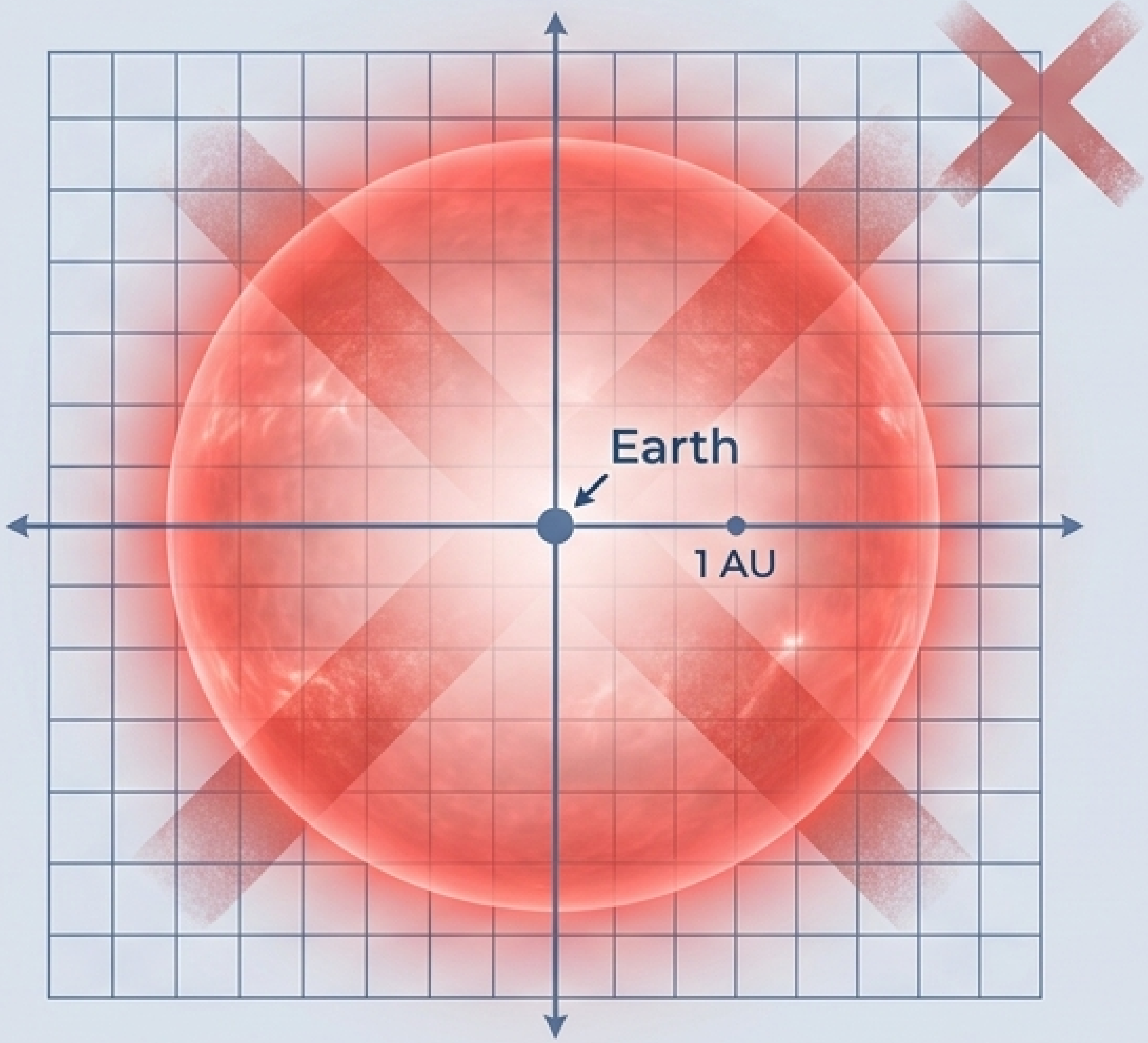
A General Theory of Prediction Failure

NotebookLM

A Flaw in the Textbook Fate of Earth

The classic forecast says the expanding Sun engulfs a stationary Earth. A 2026 paper proved the calculation right, but the premise wrong. The error wasn't in the physics. It was the assumption of the backdrop.

The Snapshot Model



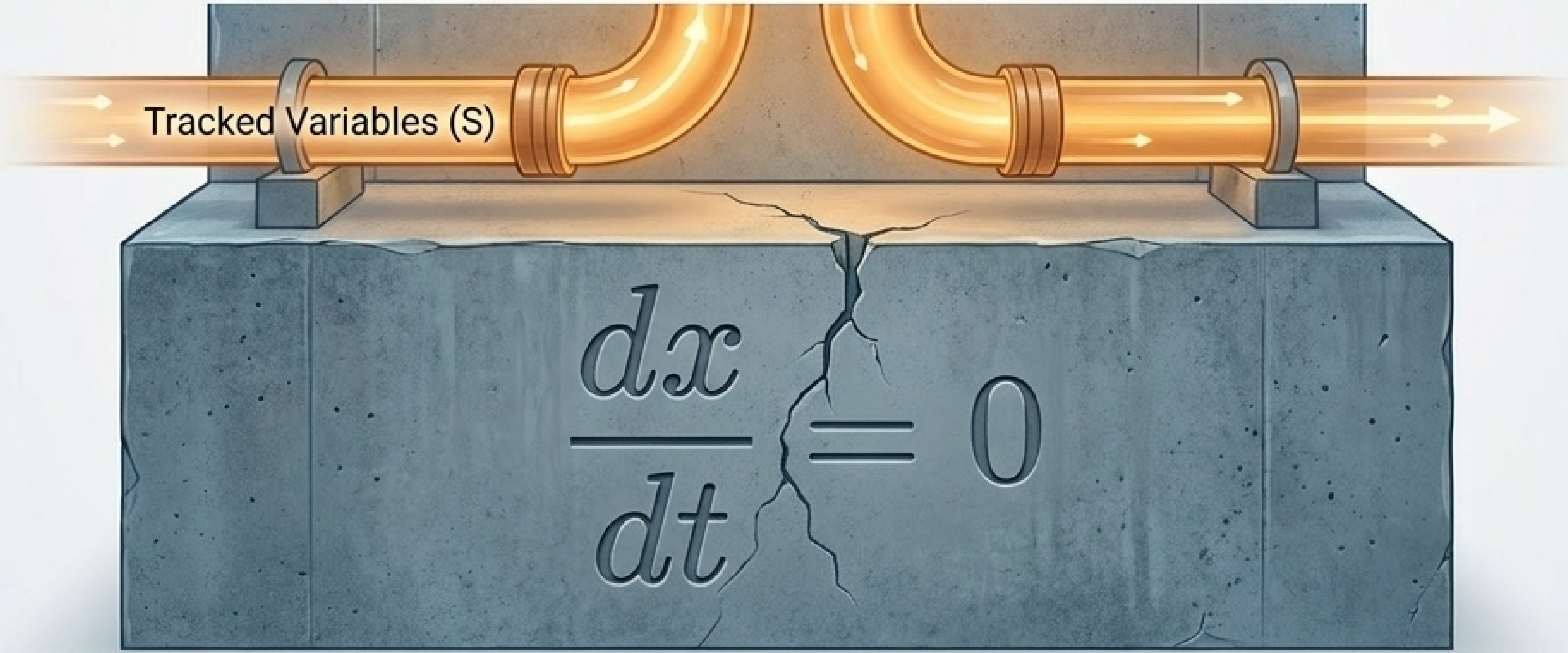
The Trajectory Model

The Calculus of Omission

A system with two coupled, moving parts gets analyzed as if it had one moving part and one fixed backdrop. The model silently sets dx/dt = 0—not by physical law, but by omission.

Frozen Process (Preliminary)

A frozen process is a trajectory assigned $\frac{dx}{dt} = 0$ — not by law, but by omission.



The Timescale Threshold

Every model must freeze variables to function. The error only occurs when the frozen quantity changes on a timescale comparable to the prediction horizon.

Frozen Process Error, Defined

A frozen-process error occurs when a quantity $x \in \Theta$ satisfies $\epsilon_x \neq 1$: the variable changes on a timescale comparable to comparable to or shorter than the prediction horizon, so freezing it introduces an error that does not vanish as the prediction is refined.

Timescale Ratio: $\epsilon_x = \tau_{\text{phenomenon}}/\tau_x$

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Forecasting as Ontological Classification

Prediction failures are rarely computational; they are ontological. Before any calculation happens, a forecaster partitions the world into things that act and things that simply are.

Prediction as Ontology

Every forecast is implicitly a classification of all quantities in the system into those that act and those that simply are. Frozen-process errors arise when that classification is made for reasons of convention, habit, or computability, rather than by checking the timescale criterion of Definition 1.1.

Parameters (Θ)

- The Stage
- What simply *is*
- Assumed fixed
- Driven by habit/convention
- Risk: Error of Omission

State Variables (S)

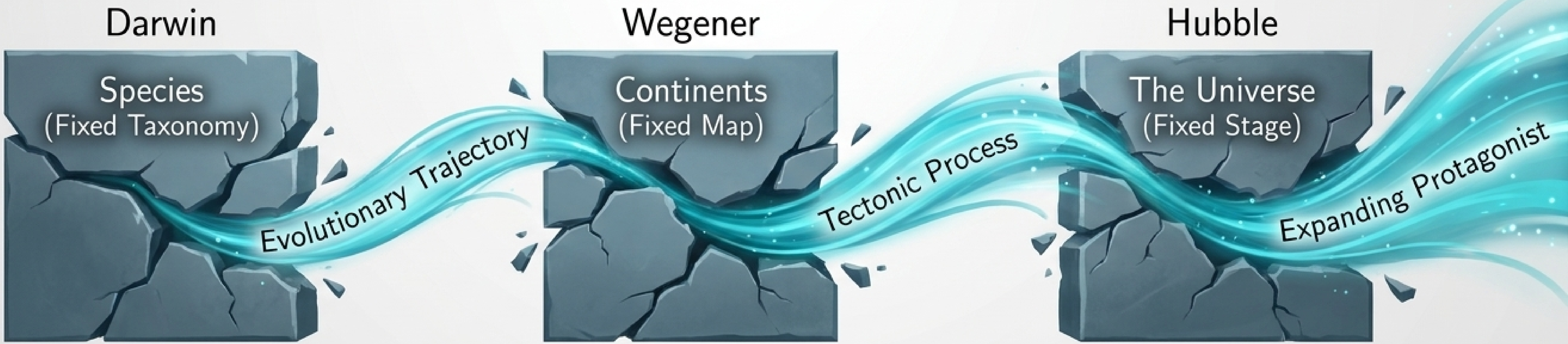
- The Protagonist
- What acts and moves
- Tracked for accuracy
- Driven by monitored dynamics

The Parameter Shift Timeline

Scientific revolutions reliably follow a single pattern: a background parameter's timescale ratio ϵ grows too large, forcing its reclassification into a dynamic trajectory.

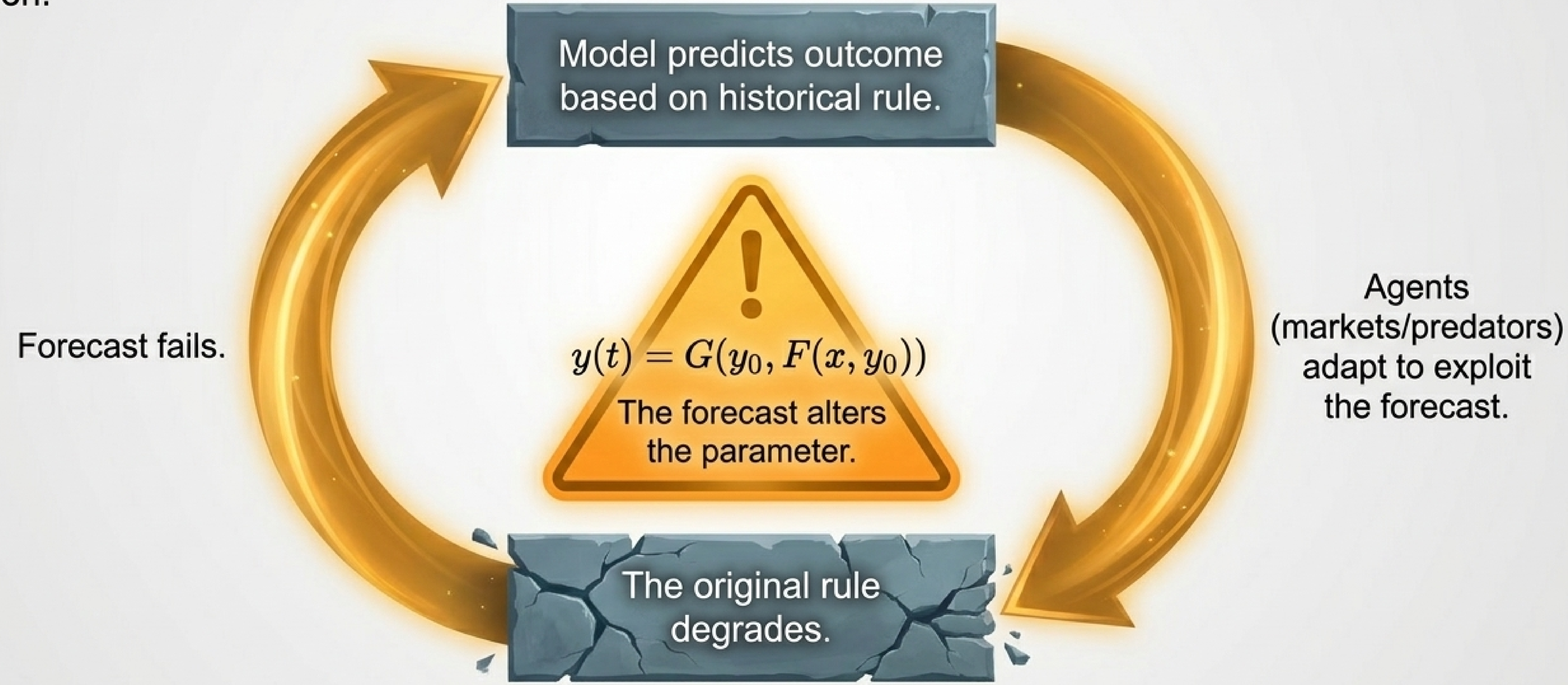
The Historical Principle

Scientific revolutions often occur not when new data arrives, but when a quantity previously classified as a fixed parameter is reclassified as a state variable—when something that *simply was* becomes something that *changes according to its dynamics*.



The Lucas Principle in Adaptive Systems

Freezing becomes exceptionally dangerous in economics, finance, and ecology. Here, the frozen quantity is a behavioral rule. The act of forecasting changes the very behavior the forecast relies upon.



The Noun Fallacy

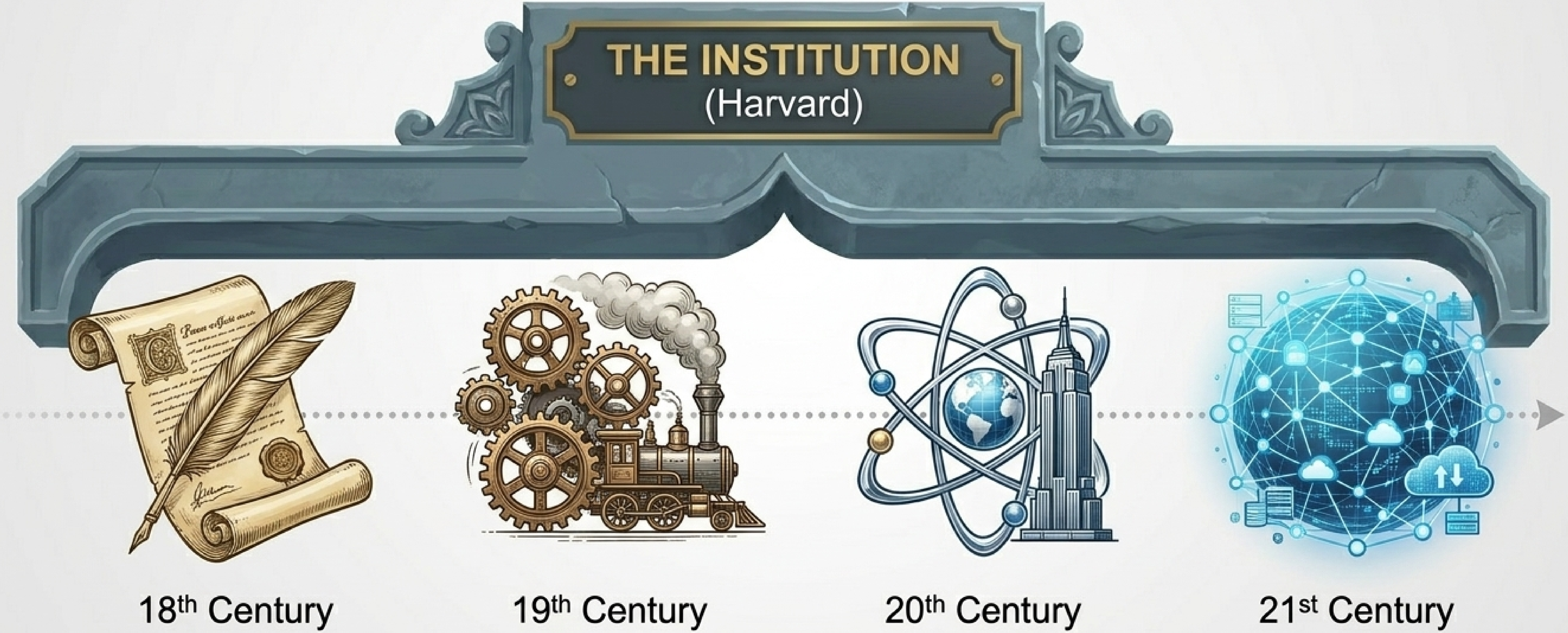
We mistake grammatical fixity for physical fixity. Nouns compress long, dynamic trajectories into single points, tricking us into assigning $dx/dt = 0$ to temporary states.

The Noun Fallacy

A noun names a trajectory stable enough to have earned a name. The frozen-process error is the mistake of inferring from the stability of the name the permanence of the trajectory.

Institutions as Compressed Trajectories

An institution named in the 18th century and operating today shares almost nothing but its name. Extrapolating its future by treating its noun as fixed commits a massive variable-freezing error.

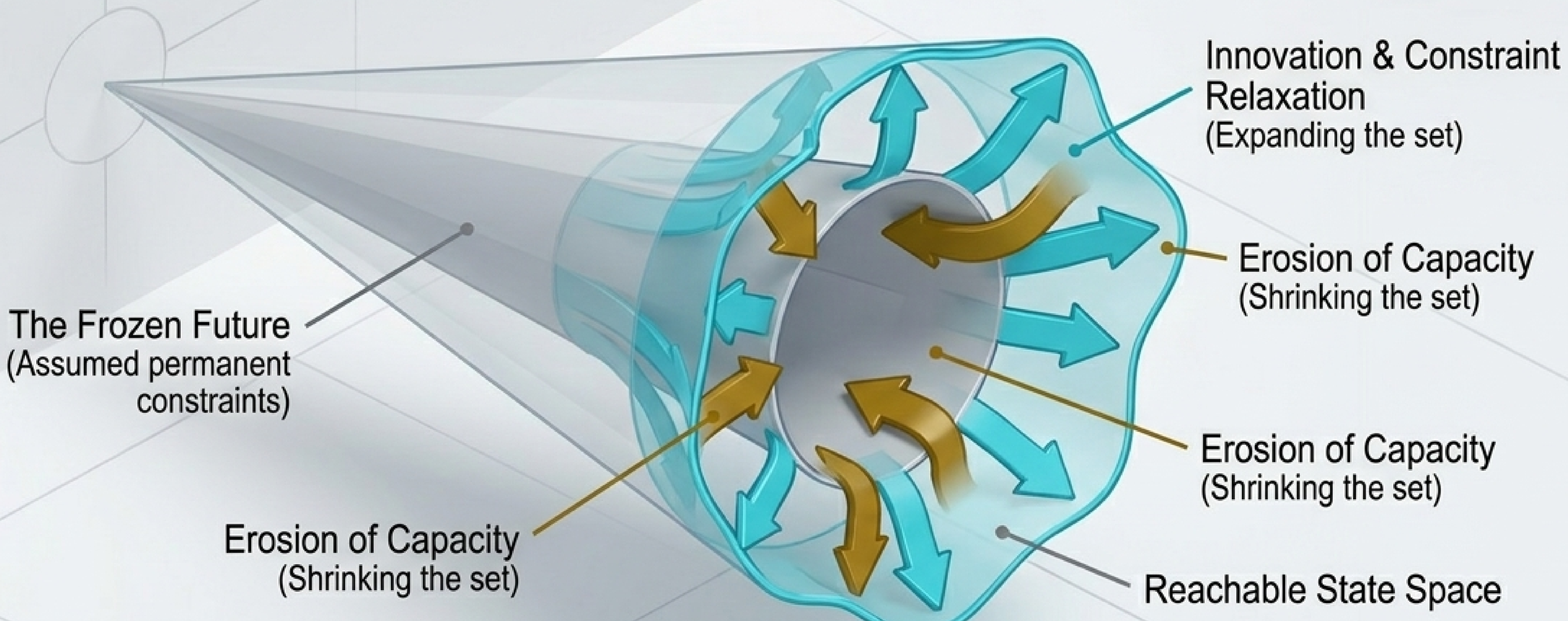


Reachability and Frozen Futures

The deepest error occurs when a model treats the space of available actions (R_T) as fixed. Malthusian models didn't just freeze food production rates; they froze the entire frontier of agricultural possibilities.

Frozen Futures

The deepest form of the frozen-process error is not a misclassified variable but a misclassified future: treating the space of available trajectories as fixed when it is itself evolving—expanding through innovation and capacity-building, contracting through erosion and depletion.



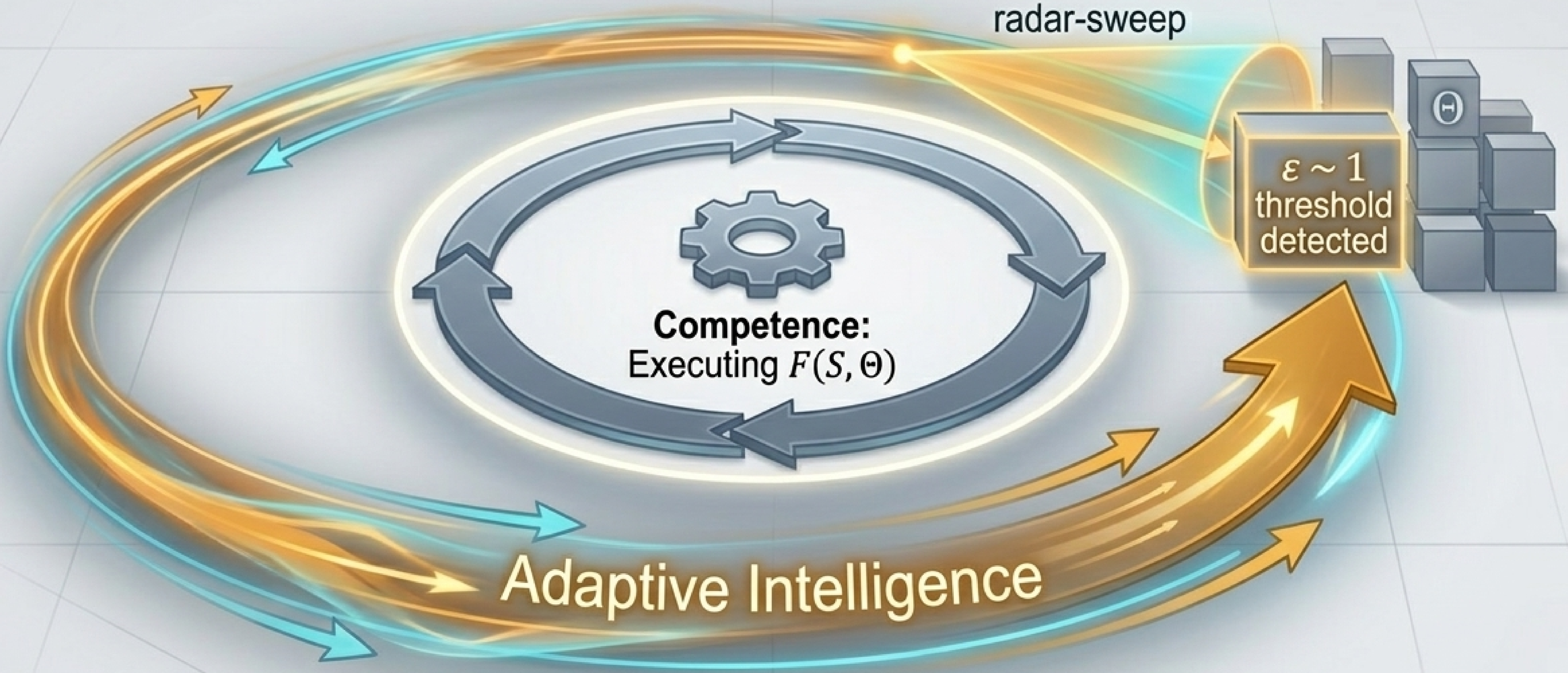
The Hierarchy of Freezing

As the complexity of the frozen process escalates, the repair requires deeper structural revisions to the model's architecture.

Level	Metric	Correction	Difficulty
Level 1: Variable Freezing	Physical quantities (e.g., Earth's orbit)	Move variable to State (S)	Local distinction revision
Level 2: Rule Freezing	Behavioral algorithms (e.g., The Lucas Critique)	Model feedback dynamics	Structural projection repair
Level 3: Possibility Freezing	Reachable futures (e.g., Civilizational constraints)	Model reachability evolution	Meta-repair of the system's capacity

Adaptive Intelligence as Ontological Repair

A basic system executes a model perfectly while the world drifts. An adaptively intelligent system constantly interrogates its background parameters, unfreezing the architecture of its model before errors accumulate.



The Final Principle

The future is often determined not by the variables we are tracking, but by the trajectories we have forgotten to notice.